Advanced Programming PROJECT

Testing:

Java JDK version : OracleOpenJDK 17.0.1
IDE used: IntelliJ
Junit test: junit-jupiter-api-5.10.0

Database Setup:
Database 1: AUTH_DB
For users and roles authentication using Hashed passwords.
Seed script: auth_db_users_auth
Tables : users_auth

Database 2:ERP_DB
For all the academic information.
Seed script: erp_db_courses, erp_db_enrollments, erp_db_grades, erp_db_instructors, erp_db_sections, erp_db_settings, erp_db_students
Tables: courses, enrollments, grades, instructors, sections, settings, students

Connection Settings:
For database 1:

```
private static final String URL="jdbc:mysql://localhost:3306/AUTH_DB";
private static final String USER="root";
private static final String PASSWORD="####";
```

For database 2:

```
private static final String URL="jdbc:mysql://localhost:3306/ERP_DB";
private static final String USER="root";
private static final String PASSWORD="####";
```

Tables:

For AUTH_DB

-> users_auth

user_id	username	role	password_hash	status	last_login
3	admin	admin	$2a$10$lW9J1I2tZsMwVeZFPoMdTezQwc10Ezq...	active	2025-11-28 18:03:43
2024001	student1	student	$2a$10$lW9J1I2tZsMwVeZFPoMdTezQwc10Ezq...	active	2025-11-28 17:54:48
3024001	instructor1	instructor	$2a$10$lW9J1I2tZsMwVeZFPoMdTezQwc10Ezq...	active	2025-11-28 17:58:42
3024002	student123	student	$2a$10$lW9J1I2tZsMwVeZFPoMdTezQwc10Ezq...	active	2025-11-26 22:24:27
3024003	inst1	instructor	$2a$10$lW9J1I2tZsMwVeZFPoMdTezQwc10Ezq...	active	NULL
3024009	student101	student	$2a$10$xStqggX05Jk9hq1Lr2Vd8edrlkKObwcBc...	active	NULL
NULL	NULL	NULL	NULL	NULL	NULL

For ERP_DB

->courses

course_id	code	title	credits
101	CSE121	Discrete Mathematics	4
103	MTH203	Multivariate Calculus	4
104	CSE201	Advanced Programming	4
111	DES121	IHCI	4
118	SSH221	SPP	4
121	CSE100	testing	4
NULL	NULL	NULL	NULL

->enrollments

enrollment_id	student_id	section_id	status	drop_deadline
2024001	2024001	1101	enrolled	2025-11-30
2024002	2024002	1101	enrolled	2025-11-30
2024006	2024006	1101	dropped	2025-11-30
2024024	2024024	1102	enrolled	2025-11-30
2024035	2024001	1102	enrolled	NULL
2024036	2024001	1105	dropped	NULL
NULL	NULL	NULL	NULL	NULL

->grades

enrollment_id	assignment	mid_sem	end_sem	quiz	score	final_grade
2024001	80.00	70.00	96.00	74.00	82.80	A
2024024	10.00	15.00	22.00	18.00	56.00	C
2024002	40.00	60.00	70.00	25.00	56.50	F

->instructors

user_id	department	designation
3024001	CSE	Professor
3024003	CSAM	assistant prof
3024012	CSAM	Assistant Professor
NULL	NULL	NULL

->sections

section_id	course_id	instructor_id	day_time	room	capacity	semester	year
1001	121	3024001	Monday 3:00PM	A007	80	summer	2025
1101	101	3024001	Monday 11AM	B003	70	Winter	2025
1102	103	3024012	Wednesday 9AM	B003	100	Winter	2025
1103	1109	3024001	12:00	B003	100	winter	0
1105	118	3024003	Thursday 1:00PM	C102	70	summer	2025
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

->settings

setting_key	setting_value
academic year	2025
current_semester	spring
maintenance_message	System under maintenance.
maintenance_mode	false
NULL	NULL

->students

user_id	roll_no	program	year	name
2024001	2241	CSE	2024	sample
2024002	2242	CSAM	2024	sample1
2024006	2246	CSB	2025	sample2
2024024	22424	CSAI	2025	sample3
2024190	2024190	CSE	2025	sample4

Acceptance Tests:

1.Login
Verify users can log in with correct credentials.
Steps: Open login screen, enter username and password, click login
Expected result: correct role dashboard loads, otherwise incorrect credentials shows error msg and return back.

2.Student views own grades
Steps: login as student1, open grades, select course, view grades for different components and final grade.
Expected result: Student sees only their own grades, No edit option.

3.Student views own enrolled courses
Steps: login as student1, open register for courses, select saved courses, view enrolled courses.
Expected result: Student sees only their own enrolled courses, No edit option if drop deadline passed or maintenance is on, in that case shows a dialog.

4.Instructor views own sections
Steps: login as instructor1, open sections, view students enrolled in that section.
Expected result: Instructor sees sections assigned to it only no other sections, No edit option.

5.Instructor updates Grades
Steps: login as instructor1, open update grades, choose section from the drop down menu, load students to display students data, update the grades and grading schema, save changes to update the grades, and compute final grade to upload final grades.
Expected result: Grades are updated in database, students can also see the changes.

6.Instructor views own sections statistics
Steps: login as instructor1, open class statistics,choose the section from the dropdown menu, view statistics for that section.
Expected result: Instructor sees sections statistics assigned to it only no other sections, No edit option, correct report display.

7.Admin create user
Steps: login as admin, open add users, add student, instructor, or new admin role.
Expected result: users created in auth db, role is correct.

8.Admin create, delete, update course
Steps: login as admin, open courses and sections tab, do the modifications in courses information.
Expected result: changes are reflected in the database, and updates acoss students and instructors is also done, duplicate entries show clear messages.

9.Admin create, delete section
Steps: login as admin, open courses and sections tab, do the modifications in sections information.
Expected result: changes are reflected in the database, and updates across students and instructors are also done, duplicate entries show clear messages.

10.Admin enables maintenance mode
Steps: login as admin, open maintenance tab, enable the maintenance mode after entering a message.
Expected result: saves the settings table with correct modification, students and instructors unable to change any data, clear ui message is visible.

11.Admin allots a instructor
Steps: login as admin, open allot instructor, enter course info, select the course to allot instructor, then next, choose the available instructors.
Expected result: Instructor allotted to that section, changes reflected in the database and instructor window.

12.Extra Tests
Checks for empty fields are added.
Checks for negative capacity is also there.

Timetable is also visible as per their roles.

Accounts to use: Admin: username:admin password:abc
Instructor: username:instructor1 password:abc
Student: username:student1 password:abc

JUnit Tests

Automated unit tests
1.usersTableTest.java
Validate user creation and username checking logic.
Test included:
1.testInsertUser()
Inserts new user
Asserts: returned user_id>0 and username exists after insertion
2.testUserNameExists()
Checks non-existing username
Asserts: returns false

2.studentsTableTest.java
Verify student data insertion
Test included:
1.insertStudent()
Inserts new student record
Queries db directly to verify
Asserts: user_id, roll_no, program, year matches

3.courseTableTest.java
Validate course creation
Tests included:
1.insertCourse()
Inserts a new course
asserts : insert operation returns true, query result contains course data(code,title,credits)

Automated Unit tests Summary:

userTableTest	2 tests	passed
studentTableTest	1 test	passed
courseTableTest	1 test	passed

To implement these tests I made modifications to the erpdatabaseconnection and authdatabaseconnection java classes. I created a method that enablesTestMode() and switches to the sample copy of these two databases. Such that we don't by mistake modify the actual database.